INCH-POUND

MIL-C-1453G
23 April 1990
SUPERSEDING
MIL-C-1453F
26 August 1970

MILITARY SPECIFICATION

CHAINS, DOG, WITH SNAPHOOK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of dog chains, each with a snaphook on one end (see 6.1).

1.2 Classification. The dog chains shall be of the following types, as specified (see 6.2).

Type I	-	106-inch kennel chain.
Type II	-	72-inch chain.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3770

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SPECIFICATIONS

FEDERAL

RR-C-271	-	Chains and Attachments, Welded and Weldless
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-676	-	Boxes Setup
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

MILITARY

MIL-S-43770/8	-	Snap Hook, Cast Body, Round Eye Swivel

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

4-1-70	-	Chain, Dog, With Snap; Type I and Type II
4-1-635	-	Markers

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-EMSS, Natick, MA 01760-5014.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 853 - Steel Wire, Carbon, for General Use
B 633 - Electrodeposited Coatings of Zinc on Iron and Steel
D 3951 - Standard Practice for Commercial Packaging

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (6.3) in accordance with 4.3.

3.2 Material. Materials shall condorm to the referenced specifications and requirements specified herein. It is encouraged that recycled material be used as long as it meets the requirements of this specification.

3.2.1 Wire, steel, carbon. The carbon steel wire shall conform to ASTM A 853.

3.2.2 Swivel. The swivel shall conform to Specification Control Drawing Number 4-1-635 or equal.

3.2.3 Snaphook. The swivel eye snaphook for the type I chain shall conform to part number M43770/8 - MIZ1 of MIL-S-43770/8. The round eye, swivel, bolt snaphook for the type II chain shall be cast of malleable iron and shall conform to North & Judd's No. 225, or equal in strength and dimensions.

3.2.4 Chain, twist link. The twist link chain shall conform to type I, grade C, class 5, style 2, size 5/32 of RR-C-271.

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3.3 Construction. The construction of the dog chain with snaphooks shall conform to the requirements shown on Drawing 4-1-70 and as specified herein. Connecting links and rings shall be constructed of material specified in 3.2.1.

3.3.1 Welding. The ends or sides of the connecting links and the ends of the rings, after being formed and assembled to twist link chain, swivel, and snaphook, shall be joined by electric welding. The surface of the parts to be welded shall be free from oxide, scale, and other foreign matter. Welds shall be sound, smooth, and free from porosity, flux, cracks, burns, and incomplete fusion.

3.4 Finish. After the removal of all burrs and sharp edges, the dog chain shall be zinc plated in accordance with type II of ASTM B 633. The zinc plating shall have a thickness of 0.00050 inches (13 micrometers).

3.5 Load test. The complete type I and type II dog chains with snaphooks shall be capable of withstanding a pull load of not less than 350 pounds without breaking or deforming, when tested as specified in 4.4.5.

3.6 Workmanship. The end item shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be

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examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristic specified in 4.4.5. Any nonconformance shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1 components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of completely assembled dog chains of one type. The sample unit shall be one dog chain. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not as specified.	101	
	Omitted from any part.	102	
Construction and workmanship	Any component missing.	103	
	Not properly assembled.	104	
	Any burrs or rough edges.	105	
	Casting contains blowholes.	106	
	Not specified number of links per chain.	107	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (cont'd)	Swivel will not rotate freely.	108	
	Rivet in snaphook loose or not peened.	109	
	Tingue not in contact with hook end of snaphook.		201
Welding	Missing where required.	110	
	Not sound; cracked or undercut	111	
	Porous		202
	Not smooth	112	
	Flux deposits not removed		203

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on Drawing 4-1-70. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of completely assembled dog chains of one type. The sample unit shall be one dog chain. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. The completely finished dog chains, with snaphooks, shall be tested for breaking strength. The assembled dog chain shall be tested on a standard tensile testing machine with a load of 350 pounds, to determine compliance with 3.5. The ring shall be blocked to prevent deformation during the test. The lot size shall be expressed in units of completely assembled dog chains of one type. The sample unit shall be one dog chain. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.0.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified

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Examine	Defect
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Contents	Number of dog chains per container is more or less than required

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A preservation. Each dog chain shall be unit packed in a snug-fitting, folding paperboard box conforming to variety I, style III, type F, class F of PPP-B-566, or a set-up paperboard box conforming to type I, variety I, class A, style 4 of PPP-B-676. The inside dimensions of each box shall approximate 5 inches in length, 3 inches in width, and 2 inches in depth. Box closure shall be secured with gummed paper tape conforming to type III, grade B of PPP-T-45.

5.1.2 Commercial preservation. Dog chains shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

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5.1.1 Level A preservation. Each dog chain shall be unit packed in a snug-fitting, folding paperboard box conforming to variety I, style III, type F, class F of PPP-B-566, or a set-up paperboard box conforming to type I, variety I, class A, style 4 of PPP-B-676. The inside dimensions of each box shall approximate 5 inches in length, 3 inches in width, and 2 inches in depth. Box closure shall be secured with gummed paper tape conforming to type III, grade B of PPP-T-45.

5.1.2 Commercial preservation. Dog chains shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Dog chains of one type only, preserved as specified in 5.1, shall be packed in a snug-fitting cleated plywood shipping container conforming to overseas type, style I of PPP-B-601. Each container shall be closed and strapped in accordance with the appendix of the box specification. The gross weight of each shipping container shall not exceed 200 pounds.

5.2.2 Level B packing. Twenty dog chains of one type only, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Chains shall be packed two in length, two in width, and five in depth within the shipping container. Inside dimensions of each shipping container shall approximate 10 inches in length, 6 inches in width, and 10 inches in depth. Approximate dimensions are furnished as a guide only. Each box shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant shipping container. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636.

5.2.3 Commercial packing. Dog chains, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), dog chains, packed as specified in 5.2.2 and 5.2.3, shall be palletized on 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, II, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 65 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The dog chains covered by this specification are used by military personnel as a stake chain or to confine dogs in kennel areas.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of chain required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
g. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Kennel
Restraint
Stake

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Review activities:

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User activities:

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Preparing activity:

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-1453G	90/04/23

3. DOCUMENT TITLE
CHAINS, DOG, WITH SNAPHOOK

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U.S. Army Natick RD&E Center	508-651-5221	256-5221
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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